

Children's use of particular linguistic devices reflect their friendship satisfaction: A Bayesian hierarchical approach

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BACKGROUND

- ❖ Language can offer insight into the nature of interpersonal relationships, including one's perceived satisfaction with their relationship.^[1]
- ❖ In adult relationships, for example:



- ❖ Despite abundant research on the formation of childhood friendships,^[8] and its developmental impacts,^[9] **the connection between children's use of specific linguistic devices and their perceptions of friendship satisfaction remains unexplored.**

PARTICIPANTS

- ❖ 30 English-speaking children (age range = 8-to-14 years; $M_{\text{age}} = 9.7$) nested in **15 friendship dyads**
 - ❖ Each child in a given dyad was matched on age (± 1 year) and expressive language abilities, and majority of the dyads were same sex friends ($n = 13$ dyads).

PROCEDURE

- ❖ Dyads engaged in a series of games and unstructured conversations, yielding **30-to-50-minutes of naturalistic language samples per dyad.**
- ❖ Children completed the *Network of Relationships Inventory-Relationship Quality Version* (NRI-RQV),^[10] and friendship satisfaction was assessed using the **satisfaction subscale of the NRI-RQV.**

LANGUAGE CODING

- ❖ Most of the linguistic devices of interest were identified via utterance-by-utterance hand coding, except for semantic alignment, which was calculated using an open-source Python program, *ALIGN*.^[11]

ANALYSES

- ❖ A **Bayesian hierarchical regression analysis** was performed using weakly informative priors, and the **leave-one-out (LOO) method** was used to compare model performance (Table 1, Figure 1).
- ❖ Due to limited data, we performed a series of **exploratory two-way interactional analyses** (Figure 2).

OBJECTIVE:

To determine whether school-aged children's use of particular linguistic devices reflect their ratings of friendship satisfaction.

Table 1. Model comparison

Model	ELPD Difference	SE Difference	Interpretation
Discourse markers removed	0.0	0.0	Reference model
Utterances removed	-0.1	1.9	Negligible drop in fit
Backchannels removed	-0.5	0.5	Minor drop in fit
Semantic alignment removed	-0.9	0.5	Minor drop in fit
Full model	-1.2	0.4	Slight overfit compared to best model
Singular pronouns removed	-1.8	3.5	Moderate decrease in fit
Plural pronouns removed	-3.9	0.8	Significant decrease in fit

ELPD = expected log pointwise predictive density; SE = standard error; a model is considered meaningfully worse if: ELPD difference $\leftarrow 2 \times \text{SE}$

Removing plural pronouns led to the largest and only statistically meaningful decrease in predictive accuracy (Table 1), indicating their unique contribution to model performance. Other predictors showed negligible (i.e., utterances, backchannels, semantic alignment) or uncertain (i.e., singular pronouns) effects.

RESULTS

Figure 1. Greater use of plural pronouns linked to higher friendship satisfaction

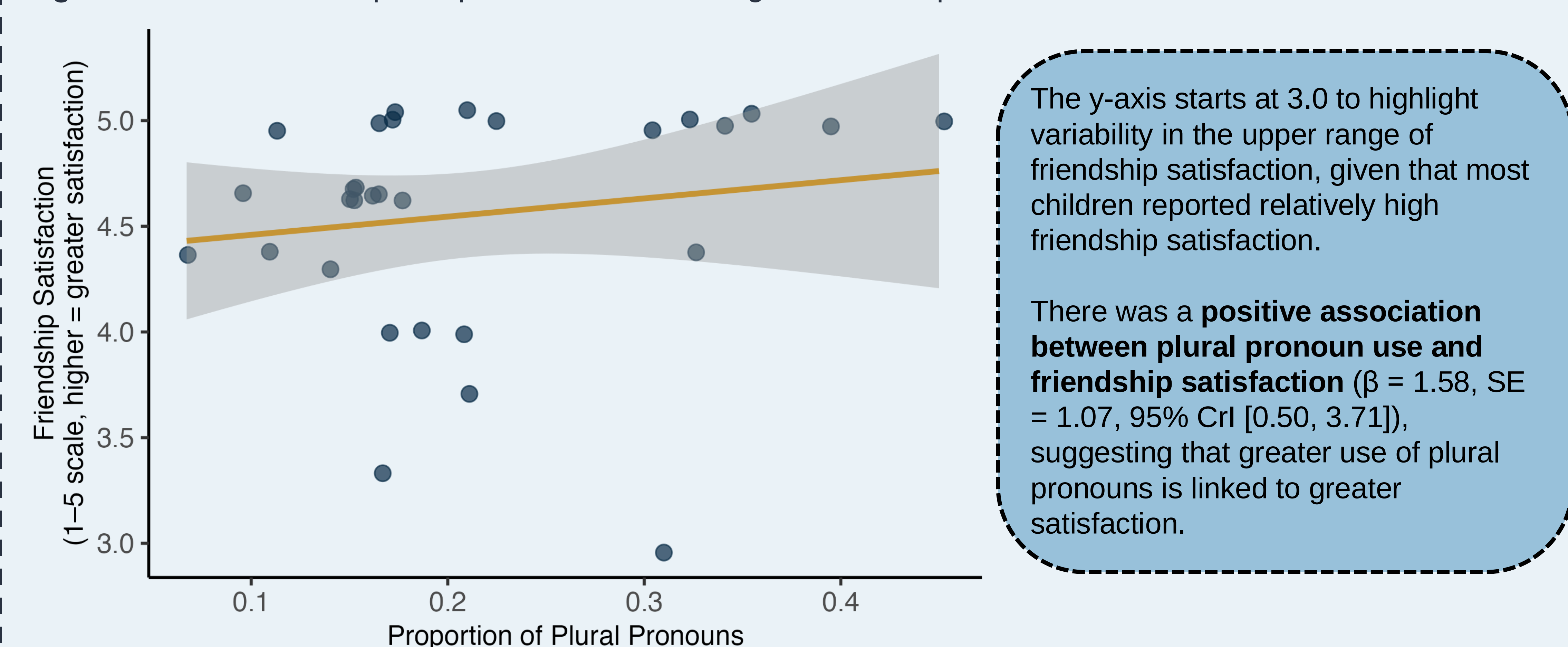
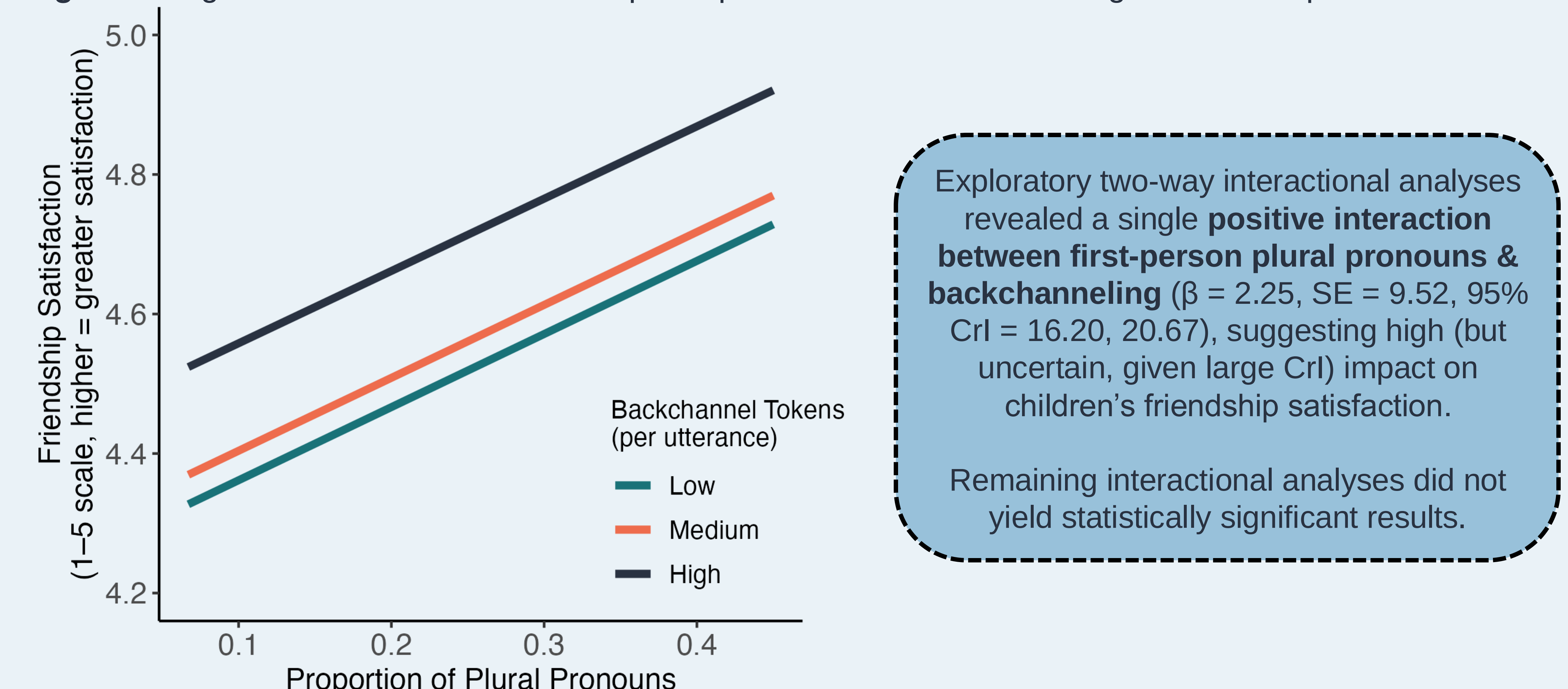


Figure 2. Significant interaction between plural pronouns & backchanneling on friendship satisfaction



DISCUSSION

- ❖ Findings suggest that children's use of **1st person plural pronouns reflect children's friendship satisfaction.**
 - ❖ It is possible that when children use pronouns like "we," they **foster** and/or **reflect a sense of inclusion** in their discourse, contributing to and/or signaling greater friendship satisfaction.
- ❖ Additionally, the **positive interaction between first-person plural pronouns and backchanneling suggests that when children backchannel, the effect of plural pronouns becomes more pronounced**, or vice versa.
 - ❖ While the directionality of this interaction is **unclear**, we speculate that when children use plural pronouns and backchannel regularly, they **create a space of active joint engagement and, thus, reflecting a satisfying relationship.**
- ❖ Overall, these findings highlight **the importance of specific linguistic choices in children's social interactions.**

REFERENCES

- [1] Tausczik & Pennebaker, 2010 [3] Slatcher et al., 2008 [5] Chung & Pennebaker, 2007 [7] Jucker, 1991 [9] Wentzel, Jablansky, & Scalise, 2018 [11] Duran et al., 2019
[2] Gonzales et al., 2010 [4] Simmons et al., 2005 [6] Rifai et al., 2022 [8] Berndt, 2002 [10] Buhrmester & Furman, 2008

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